1. Questions

Study the following information carefully and answer the following questions.

The total number of people who attended the marriage function was 400. They like to eat at least one of the two food items served, namely idly and biryani. The sum of the number of people who like only idly and only biryani together is 200. The number of people who like only idly is 20 more than the 50% of the number of people who like only biryani. The number of people who like both biryani and idly is twice that of the number of people who like only idly.

Out of the people who like only biryani, the ratio of the number of people who like chicken to mutton biryani is 21:3. Find the number of people who like only chicken biryani.

a. 101

b. 110

c. 80

d. 105

e. 45

2. Questions

Find the ratio of 20% of the number of people who like only idly to the 50% of the number of people who like only biryani.

a. 4:15

b. 3:2

c. 5:6

d. 1:1

e. 7:11

3. Questions

Find the number of people who did not like any of the two items.

a. 90

b. 40

c. 30

d. 55

e. 60

4. Questions

Out of the number of people who like both idly and biryani, 40% are males and the rest of them are females. Find the number of females who like both idly and biryani.

a. 105

b. 88

c. 120

d. 96

e. 145

5. Questions

If the number of people who like only idly is increased by 3/4th, then find the number of people who like only idly after increase.

a. 100

b. 130

c. 120

d. 90

e. 140

6. Questions

Study the following information carefully and answer the following questions.

The given table chart shows the total number of students who studied in five different colleges namely A, B, C, D and E respectively and also the ratio of the number of students who studied UG to PG in these five different colleges.

Note: The total number of students who studied = The number of students who studied UG + The number of students who studied PG.

Colleges	The total number of students who studied	The ratio of the number of students who studied UG to PG
A	1250	2:3
B	1800	11:7
C	1600	5:3
D	880	6:5
E	960	3:1

The total number of students who studied in college F is 150 less than that in college A and the sum of the number of students who studied UG in college F and A together is 5m. If the number of

students who studied PG in college F is 420, then find the value of m.

a. 185

b. 236

c. 201

d. 166

e. 205

7. Questions

In college C, 60% of the students who studied UG are male and the rest of them are female. The ratio of the number of males to females who studied PG is 2:1. Find the difference between the total number of males who studied and the total number of females who studied in that college.

a. 360

b. 450

c. 400

d. 280

e. 200

8. Questions

In College D, students studied only Commerce and Engineering in UG. If the number of UG students who studied Engineering is x/16 of the total number of UG students in College D, and the number of UG students who studied Commerce is 150, find the value of x.

a. 11

b. 5

c. 7

d. 3

e. 13

9. Questions

Find the sum of the number of students who studied PG in college B, D and E together.

a. 1240

b. 880

c. 720

d. 1100

e. 1340

10. Questions

Find the ratio between the number of students who studied UG in college B and C together to the number of students who studied PG in college D.

a. 11: 18

b. 21: 4

c. 19: 17

d. 23: 29

e. 1: 9

11. Questions

Container A contains cooking oil worth Rs. 60 per litre and container B contains cooking oil worth Rs. 80 per litre. If both the cooking oils are mixed in the ratio of 3:5 respectively, then what will be the price per litre of the mixture?

a. Rs. 85

b. Rs. 72.5

c. Rs. 97

d. Rs. 68

e. Rs. 88.5

12. Questions

Arnav invested Rs. 14500 in scheme A at the rate of 12% simple interest per annum. After 1.5 years, the total amount received in scheme A is invested in scheme B at the rate of 20% per annum for 2 years. Find the simple interest received from both schemes.

a. Rs. 9454

b. Rs. 10200

c. Rs. 3450

d. Rs. 8190

e. Rs. 11200

13. Questions

Mani and Mohan started a business with an investment of Rs. 32000 together. After 4 months Mani increased his investment by Rs. 2000 and after another 4 months, Mohan decreased his investment by Rs. 5000. At the end of the year, the ratio of the profit share of Mani to Mohan is 49:46. Find the initial investment of Mani.

a. Rs. 18000

b. Rs. 15000

c. Rs. 20000

d. Rs. 22000

e. Rs. 15600

14. Questions

Train A crosses X m long platform in 15 seconds and train B crosses 160 m long platform in 16 seconds. The ratio of the length of train A to B is 3:4 and the ratio of speed of train A to B is 4:5. Find the value of X.

a. 250

b. 160

c. 240

d. 120

e. 180

15. Questions

The ratio of the number A to B is 11:2 and the ratio of the number B to C is 1:2. The difference between (5/2) of the number C and (4/11) of the number A is 246. Find the value of B

a. 98

b. 88

c. 82

d. 26

e. 36

16. Questions

Pipes B and C together can fill a tank in 20(4/7) hours and the ratio of the efficiency of pipe B to C is 4:3. If pipe A, B and C together can fill the same tank in 11(1/13) hours, then find the time taken by pipes A and C to fill the tank alternatively.

a. 30 hours

b. 28 hours

c. 32 hours

d. 16 hours

e. 18 hours

17. Questions

The ratio of the present age of A to B is 11:5 and the age of B is 15 years more than that of C. If the difference between the age of A and C is 33 years, then find the present age of A.

a. 22 years
b. 44 years
c. 33 years
d. 66 years
e. 11 years

18. Questions

The side of the square is twice the radius of the cone. The volume and height of the cone are 4312 cm^3 and 21 cm respectively. Find the perimeter of the square.

a. 108 cm
b. 120 cm
c. 112 cm
d. 160 cm
e. 216 cm

19. Questions

Iman bought a watch and a bag. He sold the watch at a profit of 24% and bag at a loss of 8%, earning an overall profit of Rs. 80. If the ratio of the cost price of the watch to the bag is 3:5, then find the cost price of the watch.

a. Rs. 800
b. Rs. 750
c. Rs. 880
d. Rs. 720
e. Rs. 450

20. Questions

The average number of teachers working in three different schools namely A, B and C together is 210. The average number of teachers working in school A and C together is 170 and the number of teachers working in school C is 90 less than that in school B. Find the number of teachers working in school A.

a. 120
b. 210
c. 140

d. 90

e. 160

21. Questions

What value should come in the place of (?) in the following questions.

$(25)^3 * (3125)^3/(25^2)^2 = (5)^?$

a. 11

b. 14

c. 10

d. 13

e. 9

22. Questions

$? - \sqrt{2916} = 8^2 - 21\%$ of 300 + 192

a. 210

b. 247

c. 250

d. 245

e. 249

23. Questions

$$1\frac{8}{9} + 2\frac{5}{9} - 3\frac{5}{9} = ?/3$$

a. 3/5

b. 8/3

c. 9/8

d. 2/3

e. 1/2

24. Questions

60% of 300 - 15% of 420 = ?/2

a. 189

b. 245

c. 325

d. 234

e. 425

25. Questions

315 + 15 * 18 - 210 = ? * 5 - 210

a. 107

b. 215

c. 186

d. 117

e. 145

26. Questions

What approximate value should come in the place of (?) in the following questions.

32.12 * 4.98 + 126.12/18.01 = ?/1.98

a. 355

b. 334

c. 289

d. 415

e. 400

27. Questions

12.12% of 299.99 + $(1225)^{1/2}$ = (? + 4.98)/3.2

a. 198

b. 245

c. 325

d. 208

e. 200

28. Questions

210.12 * (3/7) + 17.12% of 900.12 = ?

a. 245

b. 198

c. 215

d. 243

e. 200

29. Questions

$162.12 - 90.12 + 16.12 * 2^2 = ?$

a. 224

b. 136

c. 188

d. 435

e. 302

30. Questions

$?^3 - 30.12 = 15.12 * (81.12)^{1/2} + 178.12$

a. 11

b. 8

c. 6

d. 13

e. 7

31. Questions

What value should come in the place of (?) in the following number series?

816, ?, 954, 1027, 1106, 1189

a. 883

b. 900

c. 875

d. 920

e. 890

32. Questions

1444, ?,722, 1083, 2166, 5415

a. 818

b. 980

c. 722
d. 657
e. 450

33. Questions

167, 131, 161, 137, 155, ?

a. 137
b. 88
c. 92
d. 143
e. 56

34. Questions

45, 91, 183, 367, ?, 1471

a. 735
b. 425
c. 567
d. 912
e. 818

35. Questions

455, 422, 466, ?, 477, 400

a. 413
b. 388
c. 411
d. 367
e. 421

36. Questions

Find out the wrong number in the following number series.

146, 720, 120, 840, 105, 945

a. 146
b. 120

c. 840
d. 720
e. 945

37. Questions

1255, 1280, 1315, 1345, 1385, 1430

a. 1345
b. 1315
c. 1430
d. 1280
e. 1255

38. Questions

101, 51, 54, 79.5, 161, 405

a. 51
b. 79.5
c. 161
d. 54
e. 405

39. Questions

253, 286, 324, 358, 397, 438

a. 253
b. 286
c. 358
d. 397
e. 324

40. Questions

69, 127, 218, 345, 514, 731

a. 127
b. 345
c. 514

d. 731

e. 69

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

i). $x^2 - 8x - 273 = 0$

ii). $y^2 + 32y + 252 = 0$

a. x > y

b. x ≥ y

c. x = y or relationship cannot be established

d. x < y

e. x ≤ y

42. Questions

i). $x^2 - x - 2 = 0$

ii). $2y^2 - 17y + 35 = 0$

a. x > y

b. x ≥ y

c. x = y or relationship cannot be established

d. x < y

e. x ≤ y

43. Questions

i). $x^2 + 29x + 204 = 0$

ii). $y^2 + 28y + 171 = 0$

a. x > y

b. x ≥ y

c. x = y or relationship cannot be established

d. x < y

e. x ≤ y

44. Questions

i). $x^2 - 13x + 36 = 0$

ii). $y^2 - 22y + 120 = 0$

a. x > y

b. x ≥ y

c. x = y or relationship cannot be established

d. x < y

e. x ≤ y

45. Questions

i). $x^2 - 24x + 143 = 0$

ii). $y^2 - 27y + 182 = 0$

a. x > y

b. x ≥ y

c. x = y or relationship cannot be established

d. x < y

e. x ≤ y

Explanations:

1. Questions

Let, the number of people who like only biryani = y

Given that,

(50/100)y + 20 + y = 200

0.5y + 20 + y = 200

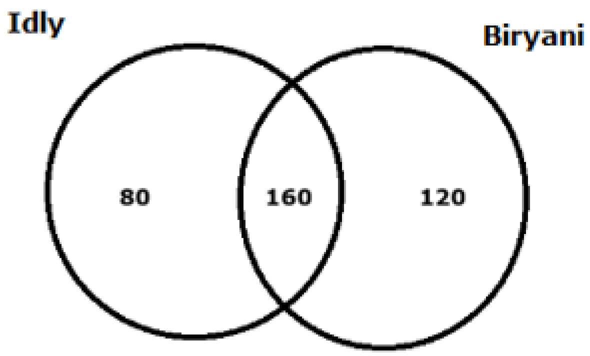
1.5y = 180

y = 120

The number of people who like only biryani = 120

The number of people who like only idly = (200 - 120) = 80

The number of people who like both idly and biryani = 80 * 2 = 160



Answer: D

The number of people who like only biryani = 120

The number of people who like only chicken biryani = 120 * 21/24 = 105

Hence, the correct answer is option D

2. Questions

Let, the number of people who like only biryani = y

Given that,

(50/100)y + 20 + y = 200

0.5y + 20 + y = 200

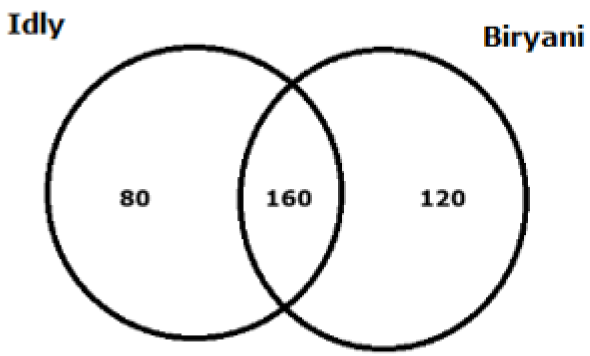
1.5y = 180

y = 120

The number of people who like only biryani = 120

The number of people who like only idly = (200 - 120) = 80

The number of people who like both idly and biryani = 80 * 2 = 160



Answer: A

The 20% of the number of people who like only idly = 80 * 20/100 = 16

The 50% of the number of people who like only biryani = 120 * 50/100 = 60

Required ratio = 16: 60 = 4: 15

Hence, the correct answer is option A

3. Questions

Let, the number of people who like only biryani = y

Given that,

(50/100)y + 20 + y = 200

0.5y + 20 + y = 200

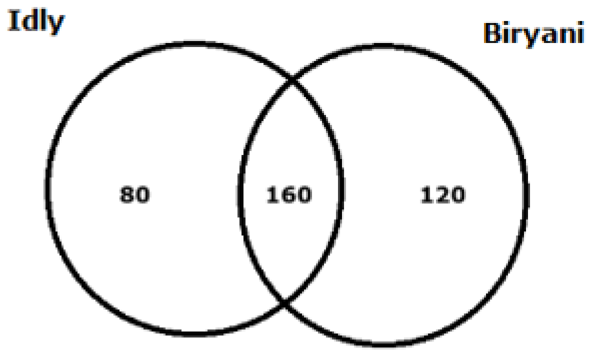
1.5y = 180

y = 120

The number of people who like only biryani = 120

The number of people who like only idly = (200 - 120) = 80

The number of people who like both idly and biryani = 80 * 2 = 160



Answer: B

The number of people who did not like any of the two items = (400 - (80 + 120 + 160)) = (400 - 360) = 40

Hence, the correct answer is option B

4. Questions

Let, the number of people who like only biryani = y

Given that,

(50/100)y + 20 + y = 200

0.5y + 20 + y = 200

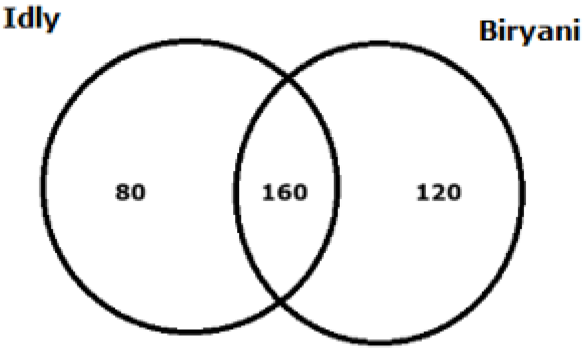
1.5y = 180

y = 120

The number of people who like only biryani = 120

The number of people who like only idly = (200 - 120) = 80

The number of people who like both idly and biryani = 80 * 2 = 160



Answer: D

The number of people who like both idly and biryani = 160

The number of females who like both idly and biryani = 160 * 60/100 = 96

Hence, the correct answer is option D

5. Questions

Let, the number of people who like only biryani = y

Given that,

(50/100)y + 20 + y = 200

0.5y + 20 + y = 200

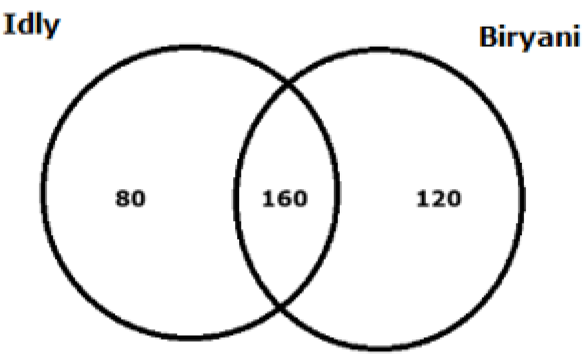
1.5y = 180

y = 120

The number of people who like only biryani = 120

The number of people who like only idly = (200 - 120) = 80

The number of people who like both idly and biryani = 80 * 2 = 160



Answer: E

The number of people who like only idly after increase = 80 + 80 * 3/4 = (80 + 60) = 140

Hence, the correct answer is option E

6. Questions

In college A:

The total number of students who studied = 1250

The number of students who studied UG = 1250 * 2/5 = 500

The number of students who studied PG = 1250 * 3/5 = 750

In college B:

The total number of students who studied = 1800

The number of students who studied UG = 1800 * 11/18 = 1100

The number of students who studied PG = 1800 * 7/18 = 700

In college C:

The total number of students who studied = 1600

The number of students who studied UG = 1600 * 5/8 = 1000

The number of students who studied PG = 1600 * 3/8 = 600

In college D:

The total number of students who studied = 880

The number of students who studied UG = 880 * 6/11 = 480

The number of students who studied PG = 880 * 5/11 = 400

In college E:

The total number of students who studied = 960

The number of students who studied UG = 960 * 3/4 = 720

The number of students who studied PG = 960 * 1/4 = 240

College	The total number of students who studied	The number of students who studied UG	The number of students who studied PG
A	1250	500	750
B	1800	1100	700
C	1600	1000	600
D	880	480	400
E	960	720	240

Answer: B

The total number of students who studied in college F = (1250 - 150) = 1100

The number of students who studied PG in college F = 420

The number of students who studied UG in college F = (1100 - 420) = 680

Given that,

680 + 500 = 5m

1180 = 5m

m = 236

Hence, the correct answer is option B

7. Questions

In college A:

The total number of students who studied = 1250

The number of students who studied UG = 1250 * 2/5 = 500

The number of students who studied PG = 1250 * 3/5 = 750

In college B:

The total number of students who studied = 1800

The number of students who studied UG = 1800 * 11/18 = 1100

The number of students who studied PG = 1800 * 7/18 = 700

In college C:

The total number of students who studied = 1600

The number of students who studied UG = 1600 * 5/8 = 1000

The number of students who studied PG = 1600 * 3/8 = 600

In college D:

The total number of students who studied = 880

The number of students who studied UG = 880 * 6/11 = 480

The number of students who studied PG = 880 * 5/11 = 400

In college E:

The total number of students who studied = 960

The number of students who studied UG = 960 * 3/4 = 720

The number of students who studied PG = 960 * 1/4 = 240

College	The total number of students who studied	The number of students who studied UG	The number of students who studied PG
A	1250	500	750
B	1800	1100	700
C	1600	1000	600
D	880	480	400
E	960	720	240

Answer: C

The number of students who studied UG in college C = 1000

The number of males studied UG in college C = 1000 * 60/100 = 600

The number of females studied UG in college C = 1000 * 40/100 = 400

The number of students who studied PG in college C = 600

The number of male students who studied PG in college C = 600 * 2/3 = 400

The number of female students who studied PG in college C = 600 * 1/3 = 200

Required difference = (600 + 400) - (400 + 200) = (1000 - 600) = 400

Hence, the correct answer is option C

8. Questions

In college A:

The total number of students who studied = 1250

The number of students who studied UG = 1250 * 2/5 = 500

The number of students who studied PG = 1250 * 3/5 = 750

In college B:

The total number of students who studied = 1800

The number of students who studied UG = 1800 * 11/18 = 1100

The number of students who studied PG = 1800 * 7/18 = 700

In college C:

The total number of students who studied = 1600

The number of students who studied UG = 1600 * 5/8 = 1000

The number of students who studied PG = 1600 * 3/8 = 600

In college D:

The total number of students who studied = 880

The number of students who studied UG = 880 * 6/11 = 480

The number of students who studied PG = 880 * 5/11 = 400

In college E:

The total number of students who studied = 960

The number of students who studied UG = 960 * 3/4 = 720

The number of students who studied PG = 960 * 1/4 = 240

College	The total number of students who studied	The number of students who studied UG	The number of students who studied PG
A	1250	500	750
B	1800	1100	700
C	1600	1000	600
D	880	480	400
E	960	720	240

Answer: A

The number of students who studied UG in college D = 480

The number of Engineering students who studied in college D = (480 - 150) = 330

Given that,

330 = x/16 * 480

330 = 30 * x

x = 11

Hence, the correct answer is option A

9. Questions

In college A:

The total number of students who studied = 1250

The number of students who studied UG = 1250 * 2/5 = 500

The number of students who studied PG = 1250 * 3/5 = 750

In college B:

The total number of students who studied = 1800

The number of students who studied UG = 1800 * 11/18 = 1100

The number of students who studied PG = 1800 * 7/18 = 700

In college C:

The total number of students who studied = 1600

The number of students who studied UG = 1600 * 5/8 = 1000

The number of students who studied PG = 1600 * 3/8 = 600

In college D:

The total number of students who studied = 880

The number of students who studied UG = 880 * 6/11 = 480

The number of students who studied PG = 880 * 5/11 = 400

In college E:

The total number of students who studied = 960

The number of students who studied UG = 960 * 3/4 = 720

The number of students who studied PG = 960 * 1/4 = 240

College	The total number of students who studied	The number of students who studied UG	The number of students who studied PG
A	1250	500	750
B	1800	1100	700
C	1600	1000	600
D	880	480	400
E	960	720	240

Answer: E

The sum of the number of students who studied PG in college B, D and E together = (700 + 400 + 240) = 1340

Hence, the correct answer is option E

10. Questions

In college A:

The total number of students who studied = 1250

The number of students who studied UG = 1250 * 2/5 = 500

The number of students who studied PG = 1250 * 3/5 = 750

In college B:

The total number of students who studied = 1800

The number of students who studied UG = 1800 * 11/18 = 1100

The number of students who studied PG = 1800 * 7/18 = 700

In college C:

The total number of students who studied = 1600

The number of students who studied UG = 1600 * 5/8 = 1000

The number of students who studied PG = 1600 * 3/8 = 600

In college D:

The total number of students who studied = 880

The number of students who studied UG = 880 * 6/11 = 480

The number of students who studied PG = 880 * 5/11 = 400

In college E:

The total number of students who studied = 960

The number of students who studied UG = 960 * 3/4 = 720

The number of students who studied PG = 960 * 1/4 = 240

College	The total number of students who studied	The number of students who studied UG	The number of students who studied PG
A	1250	500	750
B	1800	1100	700
C	1600	1000	600
D	880	480	400
E	960	720	240

Answer: B

The number of students who studied UG in college B and C together = (1100 + 1000) = 2100

The number of students who studied PG in college D = 400

Required ratio = 2100: 400 = 21: 4

Hence, the correct answer is option B

11. Questions

Answer: B

Let 3x litres of oil A mixed with 5x litres of oil B.

The total cost of the mixture 8x litres = 60 * 3x + 80 * 5x = (180x + 400x) = 580x

Price per litre mixture = 580x/8x = 72.5

Hence, the correct answer is option B

12. Questions

Answer: A

For scheme A:

SI = PNR/100

SI = 14500 * 12/100 * 1.5

SI = Rs. 2610

The total amount received in scheme A = (14500 + 2610) = Rs. 17110

For scheme B,

SI = 17110 * 20 * 2/100

SI = Rs. 6844

Required sum = (2610 + 6844) = Rs. 9454

Hence, the correct answer is option A

13. Questions

Answer: B

Let, the initial investment of Mani = Rs. x

The initial investment of Mohan = Rs. (32000 - x)

The ratio of the profit share of Mani to Mohan = ((x * 4) + (x + 2000) * 8): [(32000 - x) * 8 + (32000 - x - 5000) * 4]

= (x + 2x + 4000): (64000 - 2x + 27000 - x)

= (3x + 4000): (91000 - 3x)

(3x + 4000)/(91000 - 3x) = 49/46

138x + 184000 = 4459000 - 147x

285x = 4275000

x = 15000

The initial investment of Mani = Rs. 15000

Hence, the correct answer is option B

14. Questions

Answer: D

Let the length of trains A and B be 3b and 4b.

Let the speed of train A is 4a, so, the speed of train B is 5a.

Distance/Speed = Time

Given that,

So, (3b + X)/4a = 15

3b + X = 60a

a = (3b + X)/60 ___(1)

Also

(4b + 160)/5a = 16

4b + 160 = 80a

a = (4b + 160)/80_____(2)

Equate (1) and (2)

(3b + X)/60 = (4b +160)/80

(3b + X) * 4 = (4b + 160) * 3

12b + 4X = 12b + 480

4X = 480

X = 120

Hence, the correct answer is option D

15. Questions

Answer: C

The ratio of the number A, B and C = 11:2:4

Let, the number A, B and C be 11x, 2x and 4x respectively.

Given that,

4x * 5/2 - 11x * 4/11 = 246

10x - 4x = 246

6x = 246

x = 41

The value of B = 2x = 2 * 41 = 82

Hence, the correct answer is option C

16. Questions

Answer: C

Let, the total work = 144 units [LCM of 11(1/13), 20(4/7)]

The efficiency of pipes B and C together = 144 * (7/144) = 7 units/hour

The efficiency of pipes A, B and C together = 144 * 13/144 = 13 units/hour

The efficiency of pipe A = (13 - 7) = 6 units/hour

The efficiency of pipe B = 7 * 4/7 = 4 units/hour

The efficiency of pipe C = 7 - 4 = 3 units/hour

Required time taken = [144/(6 + 3)] * 2 = 32 hours

Hence, the correct answer is option C

17. Questions

Answer: C

Let, the present age of A and B be 11x years and 5x years respectively.

The present age of C = (5x - 15) years

Given that,

11x - 5x + 15 = 33

6x + 15 = 33

6x = 18

x = 3

The present age of A = 11x = 11 * 3 = 33 years

Hence, the correct answer is option C

18. Questions

Answer: C

The volume of the cone = V = 1/3 $\pi r^2 h$ cm^3

Given that,

4312 = (1/3) * (22/7) * r^2 * 21

196 = (1/3) * (1/7) * r^2 * 21

196 = r^2

r = 14 cm

The radius of the cone = 14 cm

The side of the square = 14 * 2 = 28 cm

The perimeter of the square = 4a cm

= 4 * 28 = 112 cm

Hence, the correct answer is option C

19. Questions

Answer: B

Let, the cost price of watch and bag is Rs. 3x and Rs. 5x respectively.

Given that,

3x * 24/100 - 5x * 8/100 = 80

72x - 40x = 8000

32x = 8000

x = 250

The cost price of the watch = 3x = 3 * 250 = Rs. 750

Hence, the correct answer is option B

20. Questions

Answer: C

According to the question,

A + B + C = 210 * 3

A + B + C = 630

Also,

A + C = 170 * 2 = 340

B = (630 - 340) = 290

The number of teachers working in school C = (290 - 90) = 200

The number of teachers working in school A = (340 - 200) = 140

Hence, the correct answer is option C

21. Questions

Answer: D

$(25)^3 * (3125)^3/(25^2)^2 = (5)^?$

$(5^2)^3 * (5^5)^3/(5^4)^2 = (5)^?$

$[5^6 * 5^{15}]/5^8 = 5^?$

$5^{21}/5^8 = 5^?$

(21 - 8) = ?

? = 13

Hence, the correct answer is option D

22. Questions

Answer: B

? - $\sqrt{2916}$ = 8^2 - 21% of 300 + 192

? - 54 = 64 - 63 + 192

? - 54 = 193

? = 193 + 54

? = 247

Hence, the correct answer is option B

23. Questions

Answer: B

$1\frac{8}{9} + 2\frac{5}{9} - 3\frac{5}{9} = ?/3$

(17/9) + (23/9) - (32/9) = ?/3

(17 + 23 - 32)/9 = ?/3

8/3 = ?

Hence, the correct answer is option B

24. Questions

Answer: D

60% of 300 - 15% of 420 = ?/2

180 - 63 = ?/2

117 = ?/2

? = 234

Hence, the correct answer is option D

25. Questions

Answer: D

315 + 15 * 18 - 210 = ? * 5 - 210

315 + 270 - 210 + 210 = ? * 5

585/5 = ?

? = 117

Hence, the correct answer is option D

26. Questions

Answer: B

32.12 * 4.98 + 126.12/18.01 = ?/1.98

32 * 5 + 126/18 = ?/2

160 + 7 = ?/2

167 * 2 = ?

? = 334

Hence, the correct answer is option B

27. Questions

Answer: D

12.12% of 299.99 + $(1225)^{1/2}$ = (? + 4.98)/3.2

12% of 300 + $(1225)^{1/2}$ = (? + 5)/3

36 + 35 = (? + 5)/3

71 * 3 = ? + 5

213 - 5 = ?

? = 208

Hence, the correct answer is option D

28. Questions

Answer: D

210.12 * (3/7) + 17.12% of 900.12 = ?

210 * 3/7 + 17% of 900 = ?

90 + 153 = ?

? = 243

Hence, the correct answer is option D

29. Questions

Answer: B

162.12 - 90.12 + 16.12 * 2^2 = ?

162 - 90 + 16 * 4 = ?

72 + 64 = ?

? = 136

Hence, the correct answer is option B

30. Questions

Answer: E

$?^3$ - 30.12 = 15.12 * $(81.12)^{1/2}$ + 178.12

$?^3$ - 30 = 15 * $(81)^{1/2}$ + 178

$?^3$ - 30 = 15 * 9 + 178

$?^3$ - 30 = 313

$?^3$ = 343

? = 7

Hence, the correct answer is option E

31. Questions

Answer: A

816 + 67 = **883**

883 + 71 = 954

954 + 73 = 1027

1027 + 79 = 1106

1106 + 83 = 1189

Hence, the correct answer is option A

32. Questions

Answer: C

1444 * 0.5 = **722**

722 * 1 = 722

722 * 1.5 = 1083

1083 * 2 = 2166

2166 * 2.5 = 5415

Hence, the correct answer is option C

33. Questions

Answer: D

167 - (6 * 6) = 131

131 + (6 * 5) = 161

161 - (6 * 4) = 137

137 + (6 * 3) = 155

155 - (6 * 2) = 143

Hence, the correct answer is option D

34. Questions

Answer: A

(45 * 2) + 1 = 91

(91 * 2) + 1 = 183

(183 * 2) + 1 = 367

(367 * 2) + 1 = **735**

(735 * 2) + 1 = 1471

Hence, the correct answer is option A

35. Questions

Answer: C

455 - 33 = 422

422 + 44 = 466

466 - 55 = **411**

411 + 66 = 477

477 - 77 = 400

Hence, the correct answer is option C

36. Questions

Answer: A

144 * 5 = 720

720/6 = 120

120 * 7 = 840

840/8 = 105

105 * 9 = 945

Hence, the correct answer is option A

37. Questions

Answer: B

1255 + 25 = 1280

1280 + 30 = **1310**

1310 + 35 = 1345

1345 + 40 = 1385

1385 + 45 = 1430

Hence, the correct answer is option B

38. Questions

Answer: D

(101 * 0.5) + 0.5 = 51

(51 * 1) + 1 = **52**

(52 * 1.5) + 1.5 = 79.5

(79.5 * 2) + 2 = 161

(161 * 2.5) + 2.5 = 405

Hence, the correct answer is option D

39. Questions

Answer: E

$16^2 - 3 = 253$

$17^2 - 3 = 286$

$18^2 - 3 = \mathbf{321}$

$19^2 - 3 = 358$

$20^2 - 3 = 397$

$21^2 - 3 = 438$

Hence, the correct answer is option E

40. Questions

Answer: E

$4^3 + 2 = \mathbf{66}$

$5^3 + 2 = 127$

$6^3 + 2 = 218$

$7^3 + 2 = 345$

$8^3 + 2 = 514$

$9^3 + 2 = 731$

Hence, the correct answer is option E

41. Questions

Answer: A

$x^2 - 8x - 273 = 0$

$x^2 - 21x + 13x - 273 = 0$

$x(x - 21) + 13(x - 21) = 0$

$(x - 21)(x + 13) = 0$

$x = 21, -13$

$y^2 + 32y + 252 = 0$

$y^2 + 18y + 14y + 252 = 0$

$y(y + 18) + 14(y + 18) = 0$

$(y + 18)(y + 14) = 0$

$y = -18, -14$

Thus, $x > y$

Hence, the correct answer is option A

42. Questions

Answer: D

$x^2 - x - 2 = 0$

$x^2 - 2x + x - 2 = 0$

$x(x - 2) + 1(x - 2) = 0$

$(x - 2)(x + 1) = 0$

$x = 2, -1$

$2y^2 - 17y + 35 = 0$

$2y^2 - 10y - 7y + 35 = 0$

$2y(y - 5) - 7(y - 5) = 0$

$(y - 5)(2y - 7) = 0$

$y = 5, 3.5$

Thus, $x < y$

Hence, the correct answer is option D

43. Questions

Answer: C

$x^2 + 29x + 204 = 0$

$x^2 + 12x + 17x + 204 = 0$

$x(x + 12) + 17(x + 12) = 0$

$(x + 12)(x + 17) = 0$

$x = -12, -17$

$y^2 + 28y + 171 = 0$

$y^2 + 19y + 9y + 171 = 0$

$y(y + 19) + 9(y + 19) = 0$

$(y + 19)(y + 9) = 0$

$y = -19, -9$

Thus, the relationship between x and y cannot be established.

Hence, the correct answer is option C

44. Questions

Answer: D

$x^2 - 13x + 36 = 0$

$x^2 - 9x - 4x + 36 = 0$

$x(x - 9) - 4(x - 9) = 0$

$(x - 9)(x - 4) = 0$

$x = 9, 4$

$y^2 - 22y + 120 = 0$

$y^2 - 10y - 12y + 120 = 0$

$y(y - 10) - 12(y - 10) = 0$

$(y - 10)(y - 12) = 0$

$y = 10, 12$

Thus, $x < y$

Hence, the correct answer is option D

45. Questions

Answer: E

$x^2 - 24x + 143 = 0$

$x^2 - 13x - 11x + 143 = 0$

$x(x - 13) - 11(x - 13) = 0$

$(x - 13)(x - 11) = 0$

$x = 13, 11$

$y^2 - 27y + 182 = 0$

$y^2 - 13y - 14y + 182 = 0$

y(y - 13) - 14(y - 13) = 0

(y - 13) (y - 14) = 0

y = 13, 14

Thus, x ≤ y

Hence, the correct answer is option E